



EAST TEXAS A&M
— UNIVERSITY —

CSCI 380.61E
WEB PROGRAMMING
AND INTERFACE DESIGN

OVERVIEW

- What do we mean by **XML** and **RSS**?
- What is RSS used for?
- XML vs JSON
- **Examples**
- Questions and Discussion

REVISION QUESTION

What does JSON stand for?

- a) Javascript Syntax Orientated Notation
- b) Just in Time Syntax Object Notion
- c) Java Orientated Notation
- d) JavaScript Object Notation

ANSWER

Answer: **d)** JavaScript Object Notation

JSON stands for JavaScript Object Notation

REVISION QUESTION

Is the following valid JSON syntax?

```
{
  "book": [
    {
      "id": "01", "language": "Java",
      "edition": "third",
      "author": "Herbert Schildt"
    },

    {
      "id": "07", "language": "C++",
      "edition": "second"
      "author": "E.Balagurusamy"
    }
  ]
}
```

ANSWER

Answer: **No**

Missing a comma after: "edition": "second"

Test in <https://jsonformatter.org/>

XML

- XML - **eXtensible Markup Language**
 - derived from Standard Generalized Markup Language (SGML)
- XML designed to store and carry the data
- XML does not perform any computation or algorithms
- XML is extensible – allows creation of self-descriptive tags
- XML has no predefined tags
- XML was developed by W3C as a public standard

XML DECLARATION

```
<?xml version = "1.0" encoding = "UTF-8"?>
<contact-info>
  <address>
    <house-no>25</house-no>
    <street>Queen Street</street>
    <postcode>Ehxxx</postcode>
  </address>
  <phone-no>07572222</phone-no>
</contact-info>
```

XML DOCUMENT STRUCTURE

- Document Prolog Section - optional
 - XML declaration
 - Document type declaration
- A correct XML document should have **only one root element**
- XML document contains two types of whitespaces
 - Significant - within the element text content
 - Insignificant - in the element tag e.g., between attributes

XML CDATA SECTION

- **CDATA** (Character Data)
 - blocks of text that are **not parsed** by the parser
 - may contain markup characters (<, >, and &)

```
<![CDATA[  
characters with markup  
]]>
```

XML NAMESPACE

- A way to avoid element name conflicts
 - specially when mixing XMLs
- Uniform Resource Identifier (URI) identifies **Namespaces** in XML

```
<f28wp:contact xmlns:f28wp = "www.f28wp.com/files">
  <f28wp:address>
    <f28wp:house-no>25</f28wp:house-no>
    <f28wp:street>Queen Street</f28wp:street>
  </f28wp:address>
  <f28wp:phone>07572222</f28wp:phone>
</f28wp:contact>
```

- Default namespace for an element -> prefixes not required

```
<contact xmlns = "www.f28wp.com/files">
  <address>
    <house-no>25</house-no>
```

XML DTDS

- **DTD (Document Type Declaration)** - a way to describe XML language precisely
- Can be used to validate an XML document
- Contains:
 - **Declarations** for elements and attributes
 - How elements are **grouped or nested**
 - **Restrictions** on the type or format of elements/attributes
- DTD can be in the document or in a separate file

```
<!DOCTYPE root-element SYSTEM "file-name">
```

XML SCHEMA - XSD

- Does the same function as DTD
- Written in XML (DTDs are driven from SGML)
- XML schema gives **more options and flexibility**
 - Allow support for namespaces
 - Define number and order of child elements
 - XML **schemas are extensible**

XML SCHEMA EXAMPLE

- In the XML file

```
<contact xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:noNamespaceSchemaLocation="f28wp.xsd">
```

- Schema

```
<?xml version = "1.0" encoding = "UTF-8"?>
<x:schema xmlns:xs = "http://www.w3.org/2001/XMLSchema">
<x:element name = "person">
  <x:complexType>
    <x:sequence>
      <x:element name="name" type="xs:string">
      <x:element name="email" type="xs:string" maxOccur="unbound">
    </x:sequence>
    <x:attribute name="age" type="xs:string" use="required"/>
  </x:complexType>
</x:element>
</x:schema>
```

XML DOM

- XML documents have a hierarchy of informational units called **nodes**
- The hierarchical structure referred to as XML Tree
- XML **Document Object Model (DOM)** is a way of describing those nodes and the hierarchical relationships between them
- XML DOM, provides API allowing to navigate, add, edit, move, or remove nodes

XML VS JSON

XML

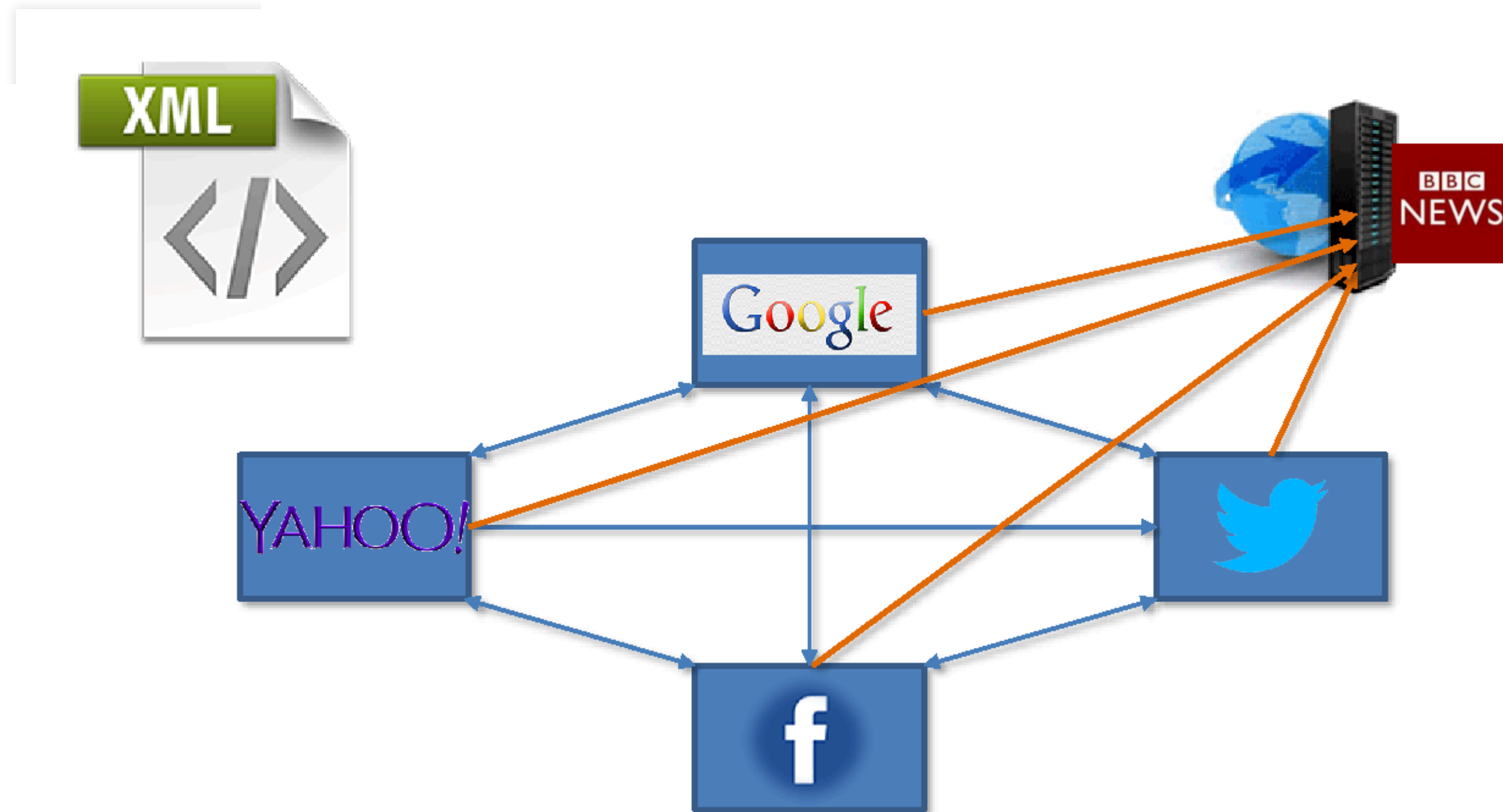
```
<employee>
  <firstName>Bob</firstName>
  <emNo>123</emNo>
</employee>
```

JSON

```
{ "employee" : {
  "firstName": "Bob",
  "emNo": 123
}
}
```

XML OR JSON?

XML OVER JSON



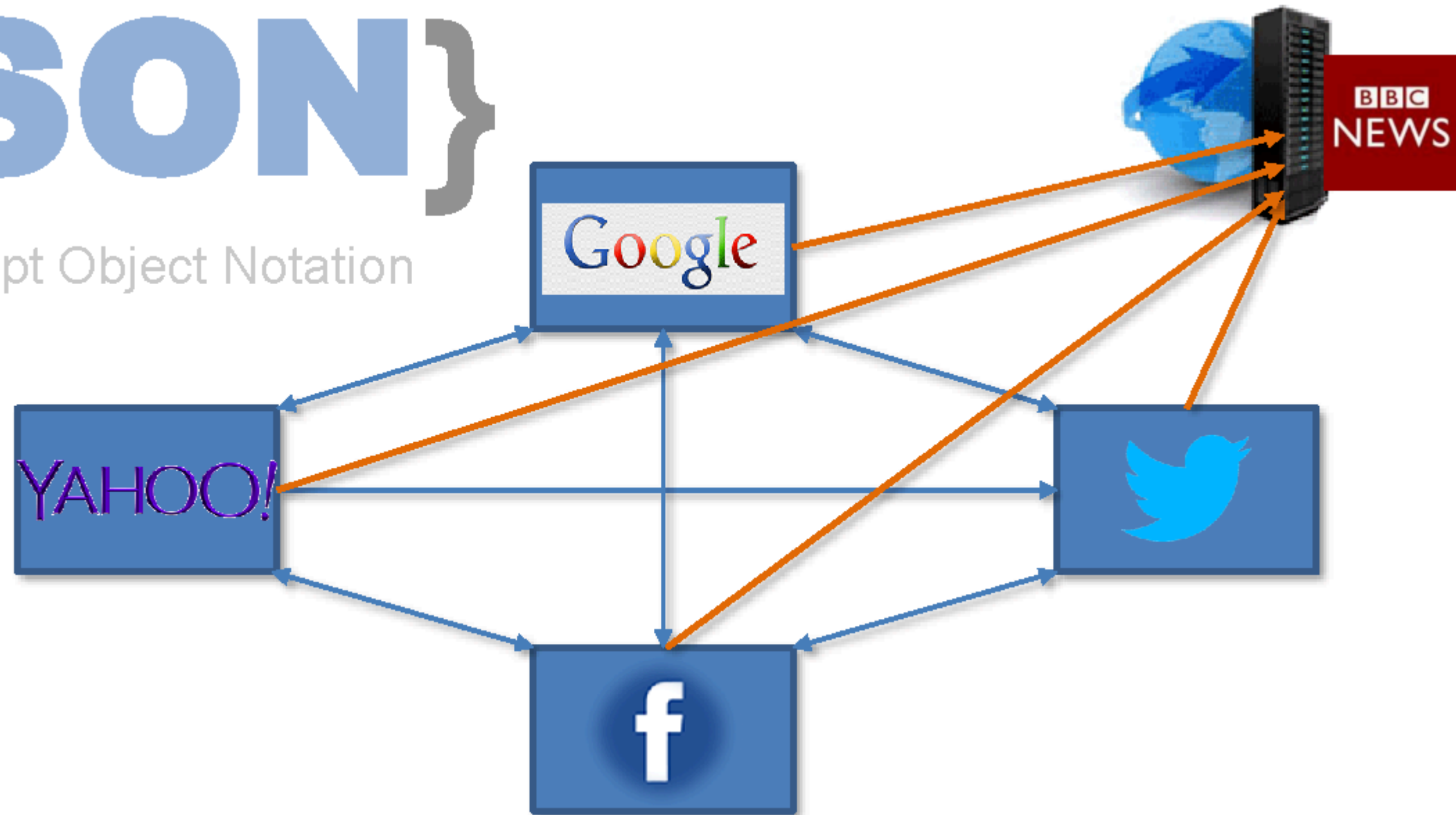
XML OVER JSON

- XML is **supported** by many more desktop applications than JSON
- Easy to take XML and apply XSLT to make XHTML or other representations
- XML is more generic and extensible
- Can use **XML Schema** to validate XML document

JSON OVER XML

{JSON}

JavaScript Object Notation



JSON OVER XML

- JSON requires less tags than XML
- JSON is **transportation-independent**, you can just bypass the XMLHttpRequest object for getting your data
- You can get JSON data from anywhere, not just your own domain (using JSONP)
- JSON is a **valid JavaScript**; better with JavaScript/Web than XML
- JSON is **easier to read** than XML

REVISION QUESTION

XML is

- a) Free and Extensible
- b) Not Extensible.
- c) A stylsheet
- d) None of these

ANSWER

Answer: **a)** Free and Extensible

REVISION QUESTION

What is CDATA? (Context of XML/XHTML)

- a) code data
- b) create data
- c) parsed character data
- d) unparsed character data (not parsed)

ANSWER

Answer: **d)** unparsed character data (not parsed)

REVISION QUESTION

XML is a Complement to

- a) GHTML
- b) HTML
- c) Xquery
- d) Xpath

ANSWER

Answer: **b)** HTML

XML (Extensible Markup Language) and HTML (Hypertext Markup Language) serve different purposes and have distinct benefits. XML is primarily used for data storage and exchange, while HTML is used for displaying and structuring web content.

REVISION QUESTION

XML is a ____ Recommendation

- a) Microsoft
- b) Sun
- c) W3C
- d) None of these

ANSWER

Answer: **c)** W3C

RSS

RSS - REALLY SIMPLE SYNDICATION

- An RSS file (feed) is a small **text** file that provides updates of a web page
- The RSS file is revised on a regular basis
- RSS is used for **information** dissemination - requires very little bandwidth
- RSS Feeds can be created manually or automatically by a software.
- RSS Feed might be registered with an RSS Feed Directory (many directories available on the web)

FAMILY

- RSS is a family of **XML**
- Typically used for **news** websites and weblogs
- Abbreviation is used to refer to the following
 - Rich Site Summary (RSS 0.91)
 - Really Simple Syndication (RSS 2.0)

SUBSCRIPTION

- RSS allows internet users to **subscribe** to RSS feeds ([details](#))
- **RSS Feeds** are read by an RSS Feed reader
 - available as standalone apps or browser extension
 - add Channels or Channel Groups

RSS ESSENTIALS

- **Essential** elements of the items in an RSS file are:
 - Title
 - Link (hyperlink)
 - Description
- RSS enables users to create information sources
- Users **subscribe** to an RSS **feed**

RSS CONTENTS (XML)

```
<rss version="2.0">
<title> NYT > Home Page </title>
<link> http://www.nytimes.com/index.html </link>
<description>
New York Times > Breaking News, World News & Multimedia
</description>
<lastBuildDate> Mon, 12 Dec 2005 05:05:00 EST </lastBuildDate>
<copyright> Copyright 2005 The New York Times Company </copyright>
<language> en-us </language>
...
```

- Examine an RSS feed and identify XML tags

RSS (Really Simple Syndication) feeds offer another way to get NYTimes.com content. Subscribe to our feeds to get the latest headlines, summaries and links back to full articles - formatted for your favorite feed reader and updated throughout the day.

NEWS

- [NYTimes.com Home Page \(U.S.\)](#)
- [World](#)
- [U.S.](#)
- [N.Y./Region](#)
- [Business](#)
- [Technology](#)
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OTHER

- [Lens Blog](#)

EXAMPLE RSS FEED

Click on RSS Feed to see the XML

Formatting when viewed in browser - why?

DISTRIBUTION VIA RSS

Information distributed by RSS, include:

- Web logs (blogs)
- Digital photo collections, such as Flickr
- Audio blogs (podcasts)
- Social bookmarking sites, such as Del.icio.us

THINGS YOU CAN DO WITH RSS

- News syndication
- **Search** for information via RSS
- Aggregate your feeds
- Display news on your website
- Display **news** on your **mobile** device
- Collect all your email in RSS
- Get RSS content through your **email**
- Track Fedex packages
- Discount air fares
- Bargains at Ebay
- Get **stock** updates ...

THINGS YOU CAN DO WITH RSS (**MORE**)

- Get the **weather** reports
- What people are saying about you, your company, your products
- Music, radio shows, TV clips
- Stay updated on someone's **schedule**
- Get cinema schedule updates
- Read your favourite comics
- Automatically backup your weblog **posts**
- Get software updates
- Get the latest bittorrent files
- **Shopping** deals at Amazon.com and others
- ...

REVISION QUESTION

What does RSS stand for?

- a) Real Script Standard
- b) Resource Scripting Source
- c) Really Simple Syndication
- d) Really Simple Script
- e) Readable Standard Source

ANSWER

Answer: **c)** Really Simple Syndication (RSS)

PARSING RSS FEEDS

- With **jQuery's** jFeed
- **XML** Compliant
 - JS Parsing and serializing XML ([details](#))

WITH JQUERY'S **BUILT-IN** XML SUPPORT

```
$.get(FEED_URL, function (data) {  
    $(data).find("entry").each(function () { // or "item" or what suits your feed  
        var el = $(this);  
  
        console.log("-----");  
        console.log("title      : " + el.find("title").text());  
        console.log("author      : " + el.find("author").text());  
        console.log("description: " + el.find("description").text());  
    });  
});
```

- 'FEED_URL' RSS Source

RSS (XML) TO JSON

- Seamlessly convert RSS feeds to JSON objects (online tools)
 - Examples
 - [apyhub](#)
 - [toptal](#)
 - [rss2json](#)

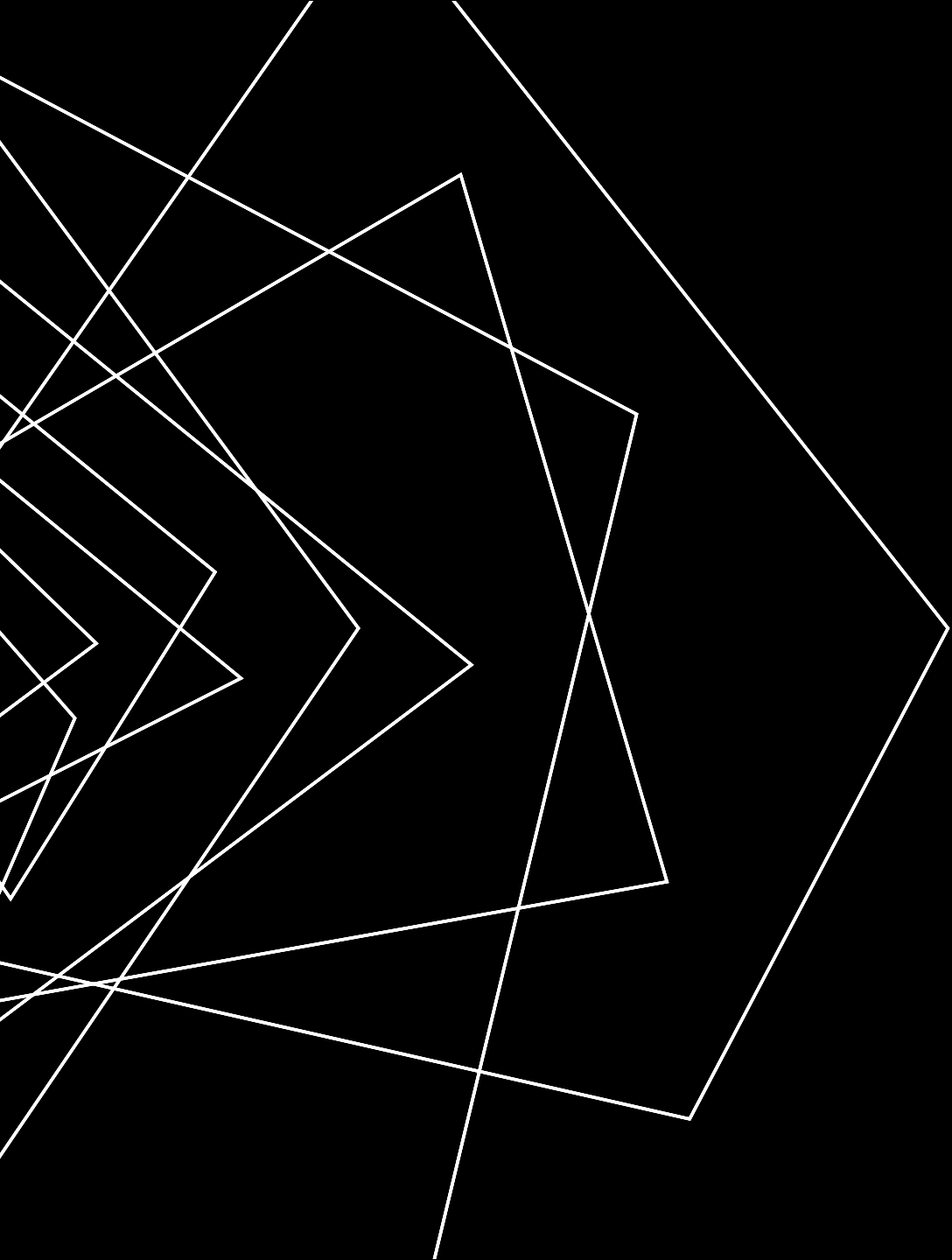
WITH JQUERY AND THE RSS2JSON

```
<script src="https://cdnjs.cloudflare.com/ajax/libs/jquery/3.7.1/jquery.min.js"></script>
<script>
$(document).ready(function(){
  $.ajax({
    url : ' https://api.rss2json.com/v1/api.json?rss_url=https%3A%2F%2Fwww.theguardian.com%2Finternat
    dataType : 'json',
    success : function (data) {
      console.log( data.feed );
      $.each(data.feed, function (i, e) { console.log("title : " + e); }); // each
    } // success
  }); // ajax
}); // ready
</script>
```

Also build url using: encodeURIComponent (FEED_URL)

SUMMARY

- Understand **XML** and **RSS**
- Whats, Whys and Hows of RSS
- Basic syntax and structure
- JSON vs XML



THANK YOU

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